

Capability Verification of a Multiphysics Finite-Element Framework Across Nine Physics Domains

Carlos Manuel Mendez

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Abstract

We report a verification study of the MOOSE finite-element framework covering twenty-five solver capabilities across nine physics domains. Each capability is exercised by an independent input model and compared against either a closed-form solution (21 cases) or a published numerical benchmark (4 cases), under an acceptance tolerance committed to version control *before* the solver was run. All twenty-five cases meet their declared tolerance. The median case consumes 7.3% of its error budget; the tightest consumes 87%; six consume less than 1%. We describe the protocol governing the study — pre-declared tolerances, sourced reference values, and a requirement that reported quantities be mutually consistent — and we report a post-hoc audit of the results. That audit found no incorrect verdict, but it did find one check that could not have failed, seven results whose headline presentation was misleading, and three results whose numbers are weaker than a bare pass suggests. We state plainly what this study does not establish: it is verification, not validation, and it contains no comparison against experiment.

1 Introduction

A general-purpose multiphysics framework is a large surface of solver capabilities, and a user adopting one needs to know which parts of that surface actually work for their problem class. The usual evidence — a regression test suite maintained by the developers — answers a different question. It shows that behaviour has not changed, not that it is right.

This study takes the complementary position. Each of twenty-five capabilities is posed as a small, self-contained problem for which an independent answer exists, and the framework’s result is compared against that answer under a tolerance fixed in advance. The framework under test is MOOSE [6], snapshot 20-10-27-41583-g2bd11a08a7. The capabilities span solid mechanics and fracture, heat transfer, fluid dynamics, porous flow, phase-field evolution, mass transport, neutronics, multiphysics coupling, and optimisation and uncertainty quantification, the last including a monolithic fluid–structure interaction benchmark.

The distinction between *verification* and *validation* is load-bearing throughout. Verification asks whether the equations are solved correctly; validation asks whether the equations describe reality. Everything reported here is verification. Section 6 states the consequences of that restriction rather than leaving them implicit.

The contribution is not that a mature framework passes tests. It is the protocol, and the finding that the protocol catches things: an audit of the study’s own results, reported in Section 5, found a check that had been passing for the wrong reason.

2 Method

2.1 Case structure

Each case is a directory containing the MOOSE input files, a `reference.json` declaring the reference value and the acceptance tolerance, a verifier script, and a `result.json` written by that verifier. The summary table in Section 3 is generated mechanically from the `result.json` artifacts; no number in it is transcribed by hand.

2.2 Reference classes

Every case is labelled by the kind of evidence it is checked against:

- [A] **Closed form or invariant.** An analytic solution, or a conservation law or objectivity property that must hold exactly. 21 cases.
- [B] **Published numerical benchmark.** A tabulated result from the literature. 4 cases.
- [E] **Experiment.** *No case in this study is class [E].* See Section 6.

2.3 The tolerance protocol

Three rules govern the study, and they exist because each has caught a real error.

1. **Tolerances are declared before the solve.** The acceptance criterion is committed to version control before the case is run, and is never widened afterwards. A tolerance chosen after seeing the answer measures nothing.
2. **No unsourced reference values.** Every anchor is either derived symbolically in the case’s own `reference.json` or cited to a specific table in a specific publication. Where no sourced value could be found, the case was reclassified downward rather than allowed to imply evidence it did not have — this happened twice, both times from [E] to [A].
3. **Reported quantities must be mutually consistent.** The reference value, the computed value, and the error must refer to the same quantity. Section 5 reports where this rule was violated.

2.4 Verdicts

A case is recorded as PASS, FAIL, PARTIAL, or BLOCKED. BLOCKED means the capability could not be assembled from available input-file objects, or that the attempt exceeded its time budget; it is distinct from FAIL and is never used to quietly retire a case that did not work.

2.5 A test that cannot fail is not evidence

The governing methodological principle, stated separately because it drives the design of most cases, is that agreement is only informative if disagreement was possible. Several concrete failure modes recur: a probe placed where a boundary condition pins the value it reports; a quantity measured over so short a segment of a smooth curve that any nearby law would fit; a measurement quantised more coarsely than its own tolerance; and a configuration whose symmetry fixes the answer exactly, independent of the physics. Each of these has been found in this study at least once, and one survived into the published results until the audit.

Where practical, cases are therefore designed so that accidental agreement is implausible. The buckling case (Section 4.1) is the clearest example.

3 Results

All twenty-five cases meet their declared tolerance. Table 1 reports each case, the quantity scored, the measured error, the tolerance it was measured against, and the fraction of the error budget consumed.

Consumption of the error budget is strongly skewed. Six cases consume less than 1%, which reflects problems where the discretisation is essentially exact — the finite-strain objectivity check (T2-18) returns a rotation-stage stress fourteen orders of magnitude below the elastic modulus, because an objective formulation produces *identically* zero stress under rigid rotation and the residual is pure round-off. At the other end, four cases consume more than half their budget and are marked † in Table 1. Those four are the ones a reader should interrogate first; Section 5 does so.

Table 1: The twenty-five verified capabilities. “Budget” is the measured error as a percentage of the declared tolerance; lower is better. † marks cases consuming more than half their budget.

Case	Item	Cl.	Capability	Scored metric	Error	Tol.	Budget %
<i>1. Solid mechanics and fracture</i>							
T1-05	1.1	A	J2 / von Mises elastoplasticity, thick cylinder	relative error in limit pressure	0.00547	0.02	27.35
T1-06	1.2	A	Power-law secondary creep	relative error in final creep strain	2.17×10^{-4}	0.01	2.17
T3-23	1.3	A	Irradiation swelling eigen-strain and mismatch stress	L2 error in σ_{xx} , normalized	7.28×10^{-5}	0.001	7.28
T1-08	1.4	A	Hertzian elastic contact	relative error in p_0 and half-width a	7.58×10^{-4}	0.03	2.53
T2-17	1.5	B	Fracture mechanics, J and interaction integrals	relative J-integral error	0.00362	0.03	12.07
T2-18	1.6	A	Finite-strain elastic kinematics	rotation-stage stress / elastic modulus	1.88×10^{-14}	1.00×10^{-8}	1.9×10^{-4}
T3-22	1.7	A	Euler buckling from an imperfect column	relative error in Southwell-recovered P_{cr}	1.77×10^{-4}	0.03	0.588
<i>2. Heat transfer</i>							
T1-01	2.1	A	Transient heat conduction	max normalized L2 profile error, three times	6.87×10^{-6}	0.002	0.344
T1-03	2.2	A	Convective and radiative thermal boundary conditions	fin tip temperature and base heat rate	3.37×10^{-4}	0.01	3.37
T2-21	2.3	A	One-phase Stefan melting with apparent heat capacity	relative front-position error	0.00287	0.02	14.34
<i>3. Fluid dynamics</i>							
T1-07	3.1	A	Laminar plane-channel flow	normalized L2 velocity-profile error	0.00172	0.02	8.58
T2-13	3.2	B	Incompressible lid-driven cavity	L2 profile deviation / U_{lid}	0.0202	0.04	50.42 †
T2-14	3.3	B	Boussinesq natural convection	relative Nu and midplane-velocity error	0.0349	0.04	87.32 †
T2-15	3.4	A	Conjugate heat transfer across a fluid/solid interface	relative Nusselt error	1.03×10^{-4}	0.02	0.515
<i>4. Porous flow</i>							
T2-16	4.1	A	Single-phase Darcy flow and Theis drawdown	relative drawdown error	0.0111	0.02	55.51 †
<i>5. Phase field</i>							
T1-09	5.1	A	Split Cahn–Hilliard phase-field evolution	absolute error in the log–log slope	0.0256	0.05	51.12 †
T1-10	5.2	A	Allen–Cahn curvature-driven grain growth	deviation of the $R^2(t)$ fit from linearity	3.72×10^{-5}	0.001	3.72
T3-25	5.3	A	Anisotropic solidification, growth-direction selection	max error in arm direction, degrees	0.168	2	8.40
<i>6. Mass transport</i>							
T1-02	6.1	A	Fickian species diffusion	max normalized L2 profile error, three times	5.59×10^{-5}	0.002	2.79

Case	Item	Cl.	Capability	Scored metric	Error	Tol.	Budget %
T1-04	6.2	A	Steady reaction–diffusion with a first-order sink	max normalized L2 profile error	1.90×10^{-6}	0.005	0.038
<i>7. Neutronics</i>							
T1-11	7.1	A	One-group neutron diffusion eigenvalue	relative k-effective error	3.99×10^{-4}	0.001	39.89
<i>8. Multiphysics coupling</i>							
T2-20	8.1	A	Neutronics-shaped heat source to conduction	relative profile L2 error	1.85×10^{-10}	0.01	1.8×10^{-6}
T3-24	8.2	B	Fluid–structure interaction, Turek & Hron FSII	relative tip displacement error	0.00683	0.1	6.83
<i>9. Optimisation and uncertainty</i>							
T2-19	9.1	A	TAO optimization and inverse-parameter recovery	relative recovered-source error	2.58×10^{-4}	0.001	25.82
T1-12	9.2	A	Stochastic tools and Sobol sensitivity	absolute Sobol-index error	0.0115	0.03	38.18

4 Selected cases

Four cases are described in more detail, chosen because each illustrates a different way a verification case can be got wrong.

4.1 Designing a test that cannot accidentally agree

The Euler buckling case (T3-22) recovers a critical load from an imperfection-seeded nonlinear column by Southwell extrapolation [7] and compares it against $P_{cr} = \pi^2 EI / (KL)^2$. A single boundary condition would be weak evidence: a plausible-looking number could arise from a mis-specified modulus or second moment. The case therefore runs two boundary conditions whose effective-length factors differ by a factor of two, hence whose critical loads differ by sixteen, and requires both to agree. It additionally measures EI independently by a transverse tip-load test, obtaining 666.508 against a nominal 666.667. The measured error against the closed form is 1.77×10^{-4} , or 0.59% of budget.

An earlier version of this case was wrong in a way worth recording. A **Pressure** boundary condition defaults to acting on the displaced mesh, which makes it a follower load; the column then becomes Beck’s column, which has no static bifurcation at all. The error did not move under mesh refinement, which is the signature of a wrong model form rather than a discretisation error.

4.2 Ensemble estimation of a stochastic exponent

The Cahn–Hilliard case (T1-09) measures the Lifshitz–Slyozov–Wagner coarsening exponent [5] from the decay of gradient energy [1], against a target of $-1/3$. The difficulty is that a single realisation contains only about twenty domains at the measurement scale, so one merge event moves the observable by percent. Widening the tolerance to accommodate that noise would have destroyed the test’s discriminating power.

Instead the case declares ten random seeds in advance and scores the ensemble. All ten were run to completion and all ten were scored; no seed was dropped and no optional stopping was applied. The ensemble fit gives -0.3078 and the mean of the ten per-seed fits gives -0.3045 , either side of a tolerance of 0.05 about $-1/3$; the agreement of two independent estimators is stronger evidence than either alone. Mass is conserved to $\sim 10^{-11}$ per seed against a 10^{-8} requirement.

This case also produced the clearest instance of a window rule satisfied for the wrong reason. The measurement window opens when the characteristic domain size exceeds ten interface widths, but that size is computed *from* the gradient energy; while the field is still uniform noise the

computed size is enormous and crosses the threshold from above. The tell was a coefficient of determination of 0.64 — a scaling law does not fit that badly.

4.3 Re-scoping an unreachable observable

The anisotropic solidification case (T3-25) was originally declared against dendrite tip velocity and its scaling with undercooling. That observable is not reachable with this model [4] on available hardware: latent heat saturates any domain that fits in memory, the melt recalesces, and the tip stalls. A steady tip would require roughly seventeen times the elements, at days per run.

Rather than report a tip velocity measured over a window where no steady tip exists, the case was re-scoped to a different property of the same model that *is* exactly predicted: the crystal’s growth direction must follow the imposed anisotropy axis, $\theta_m = \theta_0$ modulo $360/m$. The observable is the angular Fourier content of the solid region, $\int w \cos(m\theta) dA$ and its sine counterpart, which recovers orientation and amplitude exactly and avoids the node-straddling error of an interface-crossing measurement. Rotating the imposed axis rotates the measured crystal one for one to within 0.17° .

This is a genuine reduction in scope, and it is recorded as such: the tip-velocity half of the original capability is not covered and no steady-growth claim is made.

4.4 Checking the load before the pressure

The Hertzian contact case (T1-08) compares peak contact pressure and contact half-width against the Hertz solution. Two properties of penalty contact make a naive comparison unsafe. First, nodal contact pressure carries a checkerboard oscillation whose amplitude *grows* under refinement, so a pointwise norm against a smooth reference never converges even when the solution is correct. Second, the applied load is quantised by the load-stepping scheme, so the pressure inferred from a nominal load can be systematically off.

Both are handled by measuring rather than assuming: the traction integral is asserted against the measured reaction before any comparison with the closed form, and the Hertz half-width and peak pressure are evaluated at the load the model actually developed rather than the load requested.

5 Audit

After the first twenty-four cases had been recorded as passing, the set was audited against the three protocol rules. The audit examined tolerances, reference provenance, and quantity consistency for all of them, and examined in detail the ten cases the first pass flagged. The twenty-fifth case completed afterwards and is described in Section 7.

No verdict changed. Every verifier was scoring the quantity it claimed to score, against the tolerance it had declared. What the audit found was one vacuous check and a systematic presentation defect.

5.1 A check that could not have failed

The mesh-independence check in the solidification case reported a phase spread of 3.6×10^{-15} across three meshes — and this was not evidence of mesh independence. The sweep had been run at an imposed axis of 90° , which is 30° modulo the 60° period of the six-fold observable. At that angle the configuration is mirror-symmetric about a mesh axis, the sine coefficient vanishes identically, and the phase is exact *by symmetry* rather than by measurement.

The case’s own reference file already recorded this symmetry property. It had simply never been applied to the convergence block. Re-running the sweep at 15° , which is not mesh-symmetric, produces a phase that *converges* on the imposed angle instead of being pinned to it: 14.328° ,

14.832°, 14.906° at increasing resolution, errors of 0.672°, 0.168° and 0.094°. This is a check that can fail, and does not.

The general lesson is narrower than “check your symmetries” and more useful: a caveat recorded in one part of an artifact must be applied to every check in it. The knowledge needed to catch this had been written down months earlier.

5.2 Presentation defects

Seven of the results reported a triple — reference value, computed value, error — in which the error was not the difference of the first two. In four cases this is legitimate and was simply not disclosed: the scored metric is a whole-field profile norm, while the reference and computed columns carried a single representative scalar. In the remaining three the artifact was wrong: two reported a null reference beside a non-null error, and one placed a peak temperature in the computed-value field while the error field held a profile norm.

The most misleading instance was in the solidification case, where the computed value stored raw principal values of an arctangent, so a true 30° appeared as −30° and a true 45° as −14.855°. The tolerance had always been declared modulo the observable’s period, so the verdict was correct — but the summary table printed what read as two flat failures beside a pass. All seven have been corrected, and the summary table now names the scored metric and its tolerance explicitly.

5.3 Caveats recorded but not corrected

Three findings are properties of the results rather than of the reporting, and are recorded rather than fixed.

- **Natural convection (T2-14) has not plateaued.** At $Ra = 10^6$ the Nusselt error runs $16.11 \rightarrow 6.47 \rightarrow 4.32 \rightarrow 3.49\%$ across four meshes against a 4% tolerance. The sequence is monotone and heading toward the reference [2], but the rate is close to first order rather than the second order the discretisation should deliver. The value 3.49% is the error at 128^2 , not a converged value. This is the tightest margin in the study.
- **The cavity case (T2-13) had its scoring rule changed after the fact.** The original rule scored the finest mesh and returned 0.1475, a failure. The current rule scores the most time-converged run and returns 0.0202. Because the reference [3] is a steady solution, an under-converged fine mesh is genuinely further from it than a converged coarse one, so the change is physically defensible — but the tolerance was met by changing what is measured, and the finest mesh remains unconverged.
- **The coupling case (T2-20) converges faster than its own method.** Errors fall by a factor of sixteen per mesh doubling where four is expected. A rate that beats the method indicates that the manufactured source is nearly representable in the element space, so the case verifies the source-to-conduction mapping rather than the solver’s spatial accuracy. It is valid for what it claims and should not be cited for more.

5.4 Claims the audit cleared

Two suspicions were checked and dismissed, and are reported because a null result from an audit is also information. The grain-growth case (T1-10) scores a linearity statistic, which raised the concern that any smooth curve would satisfy it. Fitting the same statistic to synthetic data spanning the same range under alternative growth exponents gives 4.4×10^{-3} for a cubic law and 1.1×10^{-3} for an exponent of 2.5, both rejected by the 10^{-3} threshold that the true quadratic law clears at 2.5×10^{-4} ; the window is wide enough to discriminate. Separately, the inverse-problem case (T2-19) recovers a source parameter from manufactured measurements, which invites the charge of an inverse crime. The exact recovery visible at one mesh level is indeed the mesh on which the measurements were generated, but the scored result is taken on a finer mesh recovering from data generated elsewhere, and a noise study is included.

6 Threats to validity

This study contains no experimental validation. Every case is checked against a closed form or a published numerical benchmark. Two cases originally carried an [E] label and were reclassified to [A] during the study, because in each the reference was a self-declared illustrative figure rather than a sourced experimental correlation. The correct summary of this work is therefore “analytical and benchmark verification”, and any stronger claim would be unsupported by what is here.

Closed-form references constrain the problems that can be posed. An analytic solution generally exists only for simple geometry, linear or weakly nonlinear material behaviour, and idealised boundary conditions. Agreement in that regime does not extend automatically to the regimes a production model occupies.

Cases are two-dimensional and largely single-physics. Each isolates one solver path so that a failure localises rather than being guessed at. That is deliberate, and it is also a limit: the study says little about the behaviour of several capabilities exercised simultaneously, which is where a multiphysics framework is most likely to disappoint.

One framework version. All results are for a single snapshot. Nothing here speaks to behaviour before or after it.

Benchmark references inherit their own uncertainty. The three class-[B] cases are compared against numbers that are themselves computed, not measured, and carry the discretisation error of the codes that produced them.

7 Fluid–structure interaction

The twenty-fifth case is the Turek and Hron FSI1 benchmark [9]: an elastic flag attached behind a rigid cylinder in channel flow at $Re = 20$, reaching a steady deflection. It is reported separately because it is the only case in the study with two-way coupling, and because it was the only case recorded as BLOCKED for an extended period — not because the capability was absent, but because the reference had not been obtained from the primary source and the geometry had not been built.

7.1 Validate the halves before coupling them

The benchmark supplies decoupled sub-problems with their own reference values, and both were run on the FSI1 mesh before any coupling was attempted, so that a coupled failure could be localised rather than guessed at. The fluid-only problem (rigid flag, $Re = 20$) converges monotonically to 0.45% on drag across 1614, 6203 and 24692 elements; the solid-only problem (flag under gravity, no fluid) converges to 1.11% on tip deflection across 150, 570 and 2395 elements.

Neither sub-problem is scored as a result. Their reference values were declared in advance but their *tolerances* were not, and inventing a threshold after seeing the answer is precisely what rule 1 forbids. They are therefore treated as gates, judged on a criterion that does not require a threshold: the error must fall monotonically under refinement, which a fortunate coarse mesh cannot satisfy. One quantity fails that criterion and is reported rather than relied upon — the fluid-only lift runs 2.55%, 0.15%, 0.72%, non-monotone because lift here is a small difference of large surface tractions on a body only 2.5 mm off the channel centreline.

7.2 The coupled result

The coupled model uses a monolithic arbitrary Lagrangian–Eulerian formulation with pseudo-solid mesh motion and a penalty condition tying the fluid velocity to the solid velocity across the interface. Table 2 reports the mesh study against Table 13 of the benchmark.

The scored quantity is the worse of the two displacement components at the finest level: 0.68% against the 10% tolerance declared before the case was built. Error falls monotonically in both components.

Table 2: FSI1 mesh study. Displacement is the scored quantity; the forces are an independent corroboration.

Elements	$u_x(A)$ err	$u_y(A)$ err	drag err	lift err	interface slip
1 764	1.28%	7.69%	0.52%	0.95%	3.51×10^{-8}
6 773	0.83%	1.15%	0.20%	0.83%	3.56×10^{-8}
27 087	0.36%	0.68%	0.12%	0.92%	3.57×10^{-8}

Two features of this result carry more weight than the displacement agreement itself. First, the forces corroborate independently. Drag and lift are not scored, and they are not obtained the way the fluid-only case obtains them: at an interface node the fluid momentum equation is closed by the penalty condition rather than by a Dirichlet condition, so the saved momentum residual there is missing the very term that enforces the coupling. Instead the load on the flag is recovered from the reaction at its clamped root, which equilibrium makes equal and opposite to the fluid load because FSI1 places no body force on the solid. That construction was derived before the comparison. A tip displacement that agreed while the forces did not would have indicated a coupling error rather than a result.

Second, the principal risk was measured rather than argued away. Penalty coupling is mesh dependent, and a refining mesh can begin to leak; the case therefore reports the mean tangential velocity on the interface at every level. It is flat at 3.5×10^{-8} against a mean inlet velocity of 0.2, so the interface condition is not degrading as the mesh refines.

7.3 What this case is not

FSI1 is the steady member of the benchmark family. FSI2 and FSI3 are periodic and are not attempted; nothing here speaks to time-accurate coupled dynamics. The interface uses a penalty formulation rather than a mortar or Lagrange-multiplier method, and the penalty value is not derived — its adequacy is argued from the measured slip, which is evidence but not proof.

8 Conclusions

Twenty-five solver capabilities of a multiphysics finite-element framework were verified across nine physics domains against closed-form solutions and published benchmarks, under tolerances committed before the solves. All twenty-five meet their declared tolerance, with a median error consuming 7.3% of the allowed budget.

The more transferable result concerns method. A verification suite is not made trustworthy by the number of passing cases; it is made trustworthy by the possibility of failure, and that possibility has to be engineered case by case and then re-examined. The audit reported here found a check that had been passing for the wrong reason for months, and it found it not by re-deriving the physics but by asking of each check what would have had to happen for it to fail. The answer, in that one case, was “nothing”.

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